MENS MERCATUS

Katlego Mokoto

AI Engineer · AI Systems Architect · Agentic / LLM Developer

katlego.mokoto.km@gmail.com 065 842 4274 Johannesburg · South Africa ● AVAILABLE

AI engineer and developer — products, platforms, and agentic systems.

Portfolio · mensmercatus.online — Sound Deck, Forex Mercatus Codex, JobNet, Flutter overlay, Antigravity / local LLM stack.

ROLES

Systems architect

Full-stack engineer

.NET developer

AI architect

STACK

Languages — Python, TypeScript, JavaScript, SQL, Bash, Solidity
Frontend — React, Next.js, Flutter, Tailwind, Web Audio / DAW UIs
AI / ML — LLM orchestration, RAG, LangChain, Ollama, Agent Zero, prompt/tool agents, ChromaDB, local inferencing (Apple Silicon M4)
Backend — FastAPI, Node.js, Express, Django, WebSockets, Redis
Data — PostgreSQL, Pandas, NumPy, yfinance, OR-Tools
Cloud — Nebius, Docker, Linux, CI/CD, AWS, edge / local-first AI
Audio — Web Audio API, mic capture, stem DSP, generative synthesis
Systems — P2P/BLE mesh, agentic workflows, Cursor IDE automation

AVAILABILITY

Status — AVAILABLE
Global — Remote

FLAGSHIPS

SoundX / Sound Deck DEMO 2026

Browser DAW / AI music · Honest demo — sound stays in the browser, not a hosted session.

Original browser DAW with AI Producer (order-taking studio assistant), demo stem split, 16-step beats, mic/vocal pipelines, VST slot framework, and in-house text-to-music / stem / continuation generation.

- Generate, demo play, mic, and AI Producer on mensmercatus.online apps hub.
- Gold/black/white Mens Mercatus product UI; no third-party music-site dependency.

`Web Audio · Agents · DSP · TypeScript/JS`

JobNet DEMO 2026

Career automation · Draft-only — live read-only shortlist demo; never applies or sends mail.

Weekday pipeline: noon multi-board scrape; CV tailor + AI Hawk draft fills with human confirm before submit; Gmail reply watch + SMS/email notify.

- Noon scrape across top AI employers + early-stage trackers.
- 3 AM CV tailor + draft fills; human-in-the-loop gates before any submit.

`Automation · Agents · Gmail · macOS shell`

Mercatus Cartel DEMO 2026

Event-mode pack · Static event-mode pack; Assist LIVE path with DEMO fallback if VPS down.

Screens, schematics, and compliance gates on the site. Assist LIVE: same-origin /api/cartel/ → VPS robot.

- Event-mode product pack with architecture diagrams and compliance gates.
- Human-in-the-loop assist path with honest DEMO labels.

`Event systems · Compliance · Assist · UI systems`

Forex Mercatus Codex DEMO

2026

Python desk :8790 · Simulated paper research only — not a brokerage.

~87MB Python + Highcharts desk for FX workflows; Chart, Multi, Whales, Flow, Bot. Hosted on VPS as live paper research.

- Inventory and load-path optimization for faster cold start and clearer static serving.
- Market data fused with interactive analysis desks.

`Python · Highcharts · yfinance · Paper only`

SELECTED WORK

Project Antigravity

Mar 2026 – Present

AI Systems Architect & Founder

- Sovereign, decentralized AI OS for autonomous task execution and specialized agentic workflows.
- Terminal-first AI tooling on M4 with Ollama; Nebius for scalable training/deploy of custom models; Agent Zero + Cursor orchestration.

`Ollama · Agent Zero · Nebius · LLM Ops`

Mens Mercatus Ecosystem

Ongoing · mensmercatus.online

Full-stack & AI trading systems

- Telegram AI trading assistant: RAG + live financial feeds for XAUUSD / cyclical models (SMT, Gann frameworks) with institutional-order context.
- Brand, UI/UX, and static portfolio architecture for the Mens Mercatus digital ecosystem.

`Python · LangChain · ChromaDB · Next.js`

NARRATIVE

- Hands-on builder of agentic products: copilots that execute studio/job/market actions with explicit user confirmation gates.
- Comfortable in Python and TypeScript full-stack; fast iteration with Cursor, local models, and cloud GPUs when needed.
- Background in fleet/logistics optimization (OR-Tools) and diesel/mechanical operations — strong systems thinking under real constraints.

Portfolio: mensmercatus.online · Homelake / Randfontein · Open to remote AI eng roles